

Logos-SIE

Synthetic Information Ecosystem for Truth Discovery and Retrieval

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Abstract

Logos-SIE (Synthetic Information Ecosystem for Truth Discovery and Retrieval) is a synthetic benchmark generation framework designed to create controlled information environments for evaluating retrieval systems, graph reasoning systems, trust estimation algorithms, and truth discovery methods.

Unlike traditional benchmarks that provide only static facts, Logos-SIE explicitly models the process through which information is generated, observed, interpreted, and transformed into beliefs. The framework generates a hidden ground-truth world, simulates information sources with varying reliability characteristics, organizes those sources into communities, generates observations containing uncertainty, and finally derives beliefs from observed evidence.

The resulting dataset exposes every stage of information generation, enabling reproducible experimentation in retrieval, source attribution, trust modeling, graph traversal, multi-hop reasoning, and truth discovery.

This document describes the architecture, generation process, dataset composition, topology characteristics, benchmark statistics, and intended research applications of the first public Logos-SIE release.

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1 Executive Summary

Logos-SIE is a synthetic information ecosystem generator designed to model the complete lifecycle of information.

Rather than releasing only a knowledge graph, Logos-SIE generates multiple interconnected layers:

1. Hidden ground-truth facts
2. Information sources
3. Source communities
4. Observations
5. Beliefs

The benchmark release described in this report contains:

- 1,000 entities
- 5,000 facts
- 4,236 unique relationships
- 100 information sources
- 3 source communities
- 500,000 observations
- 500,000 beliefs
- Estimated reasoning depth of 42
- Effective branching factor of 4.406

The objective is not to simulate any specific real-world domain. Instead, Logos-SIE provides a controllable synthetic environment in which every fact, observation, belief, and source characteristic is known.

This enables rigorous evaluation of systems that must reason under uncertainty and conflicting evidence.

2 Motivation

Most retrieval and reasoning benchmarks evaluate systems using static corpora or fixed knowledge graphs.

These benchmarks typically expose only the final information available to a model.

In real information ecosystems, however, information emerges through a sequence of processes:

1. Events occur.
2. Facts describe those events.

3. Sources observe facts.
4. Observations contain noise and uncertainty.
5. Sources form beliefs.
6. Communities influence information propagation.

This distinction becomes important when evaluating:

- Retrieval systems
- Truth discovery algorithms
- Source attribution systems
- Trust estimation methods
- Graph reasoning systems
- Multi-hop reasoning models

Logos-SIE was developed to expose the complete information-generation process rather than only the final outputs.

3 Framework Overview

The framework generates a synthetic information ecosystem through five stages.

3.1 Truth Graph Generation

The first stage creates a hidden ground-truth world represented as a directed graph.

Entities become graph nodes and facts become graph edges.

Multiple topological motifs are deliberately introduced:

- DAG structures
- Hierarchical structures
- Local SCCs
- Hub nodes
- Bridge edges
- Conflict structures
- Random connections

This avoids the collapse into trivial ring graphs and promotes diverse reasoning paths.

3.2 Source Generation

Information sources are generated independently of the graph.

Each source receives attributes that influence observation quality and belief formation.

3.3 Community Formation

Sources are grouped into communities based on reliability characteristics.

The current implementation generates:

- Reliable communities
- Mixed communities
- Skeptical communities

Each community is assigned a cohesion score.

3.4 Observation Generation

Sources observe portions of the hidden truth graph.

Observations contain:

- Measured values
- Uncertainty estimates
- Temporal information
- Ontology tags
- Causal chains

Observed values may differ from the underlying truth value.

3.5 Belief Formation

Beliefs are generated from observations.

Beliefs represent the source's interpretation of observed evidence.

Each belief retains links to:

- Source identifiers
- Fact identifiers
- Evidence identifiers
- Ontology tags
- Causal chains

This allows downstream evaluation of source reliability and truth-discovery systems.

4 Generation Pipeline

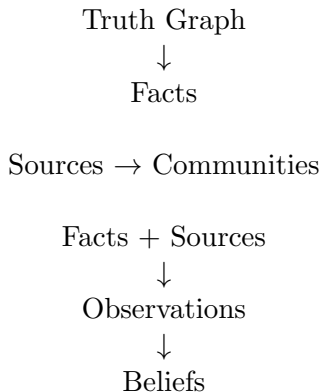


Figure 1: Logos-SIE generation pipeline.

5 Dataset Structure

The benchmark release contains the following artifacts.

File	Description
<code>sources.json</code>	Generated information sources
<code>communities.json</code>	Source community assignments
<code>observations.json</code>	Observed facts and measurements
<code>beliefs.json</code>	Beliefs derived from observations
<code>knowledge_base.json</code>	Combined ecosystem representation
<code>benchmark_statistics.json</code>	Topology metrics
<code>reasoning_examples.json</code>	Multi-hop reasoning chains

Table 1: Dataset artifacts.

6 Topology Validation

The generated benchmark was validated using graph-theoretic metrics.

6.1 Benchmark Statistics

Metric	Value
Entities	1000
Facts	5000
Relationships	4236
Unique Subject-Object Edges	4236
Unique Triples	4984
Duplicate Edge Ratio	0.153
Duplicate Triple Ratio	0.003
Number of SCCs	224
Largest SCC Size	82
Largest SCC Fraction	0.082
Average In Degree	4.236
Average Out Degree	4.236
Average Shortest Path Length	3.669
Graph Diameter	7
Longest Acyclic Chain	34
Estimated Reasoning Depth	42
Effective Branching Factor	4.406
Bridge Density	0.020
Top Hub Degree Share	0.024
Predicate Entropy	4.607
Benchmark Health	HEALTHY

Table 2: Topology statistics.

7 Interpretation of Topology Metrics

Several observations can be made from the benchmark statistics.

1. The graph avoids collapse into a giant SCC.
2. Duplicate relationships remain low.
3. Multiple reasoning pathways exist.
4. Branching behavior is present.
5. Predicate diversity remains high.
6. Hub concentration remains limited.
7. Community structures emerge naturally.

These properties collectively indicate a benchmark suitable for retrieval and graph reasoning evaluation.

8 Community Analysis

Three source communities are generated.

8.1 Reliable Community

Sources exhibiting high reliability characteristics.

Cohesion Score: 0.84

8.2 Mixed Community

Sources exhibiting mixed reliability behavior.

Cohesion Score: 0.73

8.3 Skeptical Community

Sources exhibiting lower reliability characteristics.

Cohesion Score: 0.62

These communities provide a basis for evaluating source trust estimation methods.

9 Knowledge Representation

Each belief includes:

- Source identifier
- Fact identifier
- Confidence score
- Valence score
- Evidence references
- Ontology tags
- Temporal windows
- Ground-truth quality estimates

Each observation includes:

- Truth value
- Observed value
- Measurement uncertainty
- Timestamp
- Ontology annotations
- Causal chains

This structure supports truth discovery experiments where systems must infer latent truth from noisy evidence.

10 Supported Research Tasks

The benchmark is designed to support:

- Truth discovery
- Retrieval evaluation
- Multi-hop reasoning
- Graph traversal
- Community detection
- Trust estimation
- Source attribution
- Belief aggregation
- Evidence tracking
- Knowledge graph analytics

11 Limitations

The current release remains synthetic.

Several limitations should be acknowledged:

- No natural language documents are generated.
- Sources do not currently influence one another.
- Belief propagation is limited.
- Temporal evolution remains simplified.
- Community interactions are static.

Future versions will expand these capabilities.

12 Future Work

Planned extensions include:

- Dynamic source interactions
- Temporal world evolution
- Community influence propagation
- Contradictory narratives
- Natural language document generation

- Synthetic misinformation campaigns
- Adversarial retrieval benchmarks
- Ground-truth hidden-world evaluation tracks

13 Conclusion

Logos-SIE provides a synthetic information ecosystem rather than a traditional benchmark dataset.

By exposing ground truth, observations, beliefs, sources, and communities simultaneously, the framework enables controlled experimentation across multiple research areas.

The current release demonstrates that large-scale synthetic ecosystems can be generated with measurable topology characteristics, substantial reasoning depth, diverse relationship structures, and explicit source-level provenance.

The dataset is intended as a foundation for future work in retrieval, trust estimation, graph reasoning, and truth discovery.